

Linux Commands Cheat Sheet

With examples · from basic commands towards cybersecurity

How to use this sheet: read a section, then open a Linux terminal and type every example yourself. Lines starting with \$ are what you type; lines below them are sample output (yours will differ). Do not memorize the commands - practice them.

1 Navigation Commands

Command	Use	Example
<code>pwd</code>	Show current directory	<code>pwd</code>
<code>ls</code>	List files	<code>ls</code>
<code>ls -l</code>	Detailed list	<code>ls -l</code>
<code>ls -a</code>	Show hidden files	<code>ls -a</code>
<code>cd</code>	Change directory	<code>cd /home</code>
<code>cd ..</code>	Go one level back	<code>cd ..</code>
<code>cd ~</code>	Go to home directory	<code>cd ~</code>

Example (try it in your terminal):

```
$ pwd
/home/student
$ ls -la
drwxr-xr-x 3 student student 4096 Sep 30 09:10 .
-rw-r--r-- 1 student student 220 Sep 30 09:10 .bash_logout
drwxr-xr-x 2 student student 4096 Sep 30 09:12 Documents
$ cd /home
$ cd ..
$ pwd
/
```

2 File & Directory Commands

Command	Use	Example
<code>mkdir</code>	Create directory	<code>mkdir test</code>
<code>rmdir</code>	Remove empty directory	<code>rmdir test</code>
<code>touch</code>	Create file	<code>touch file.txt</code>
<code>cat</code>	Display file	<code>cat file.txt</code>
<code>cp</code>	Copy file	<code>cp file.txt backup.txt</code>
<code>mv</code>	Move / rename	<code>mv file.txt new.txt</code>
<code>rm</code>	Delete file	<code>rm new.txt</code>

Example (try it in your terminal):

```
$ mkdir cybersecurity
$ cd cybersecurity
$ touch notes.txt
$ echo "Linux Practice" > notes.txt
$ cat notes.txt
Linux Practice
$ cp notes.txt backup.txt
$ mv backup.txt old_notes.txt
$ rm old_notes.txt
```

Security note: rm deletes permanently (there is no recycle bin). Double-check the file name before pressing Enter. Use rm -r only for folders, and never run rm -rf / .

3 User Commands

Command	Use	Example
<code>whoami</code>	Current username	<code>whoami</code>
<code>id</code>	User and group IDs	<code>id</code>
<code>who</code>	Logged-in users	<code>who</code>
<code>w</code>	Users + activity	<code>w</code>
<code>passwd</code>	Change password	<code>passwd</code>

Example (try it in your terminal):

```
$ whoami
student
$ id
uid=1000(student) gid=1000(student) groups=1000(student),27(sudo)
$ who
student  tty7    2026-09-30 09:05
```

4 Permission Commands

ls -l shows permissions like -rwxr-xr-x (owner, group, others). chmod changes permissions, chown changes the owner.

Command	Use	Example
<code>ls -l</code>	View permissions	<code>ls -l secret.txt</code>
<code>chmod</code>	Change permissions	<code>chmod 755 script.sh</code>
<code>chown</code>	Change file owner	<code>chown user:user file.txt</code>

Example (try it in your terminal):

```
$ touch secret.txt
$ chmod 600 secret.txt
$ ls -l secret.txt
-rw----- 1 student student 0 Sep 30 09:20 secret.txt
```

Permission numbers:

```
r = 4   w = 2   x = 1
755 = rwxr-xr-x (owner: all, others: read + run)
644 = rw-r--r-- (owner: read/write, others: read)
600 = rw----- (only the owner can read/write)
```

Security note: Permissions decide who can read, change or run a file. Files holding secrets (keys, passwords) should be 600 so only their owner can access them.

5 System Information

Command	Shows	Example
<code>uname -a</code>	Kernel / system information	<code>uname -a</code>
<code>hostname</code>	Computer name	<code>hostname</code>
<code>cat /etc/os-release</code>	Linux distribution / version	<code>cat /etc/os-release</code>
<code>uptime</code>	System running time	<code>uptime</code>
<code>free -h</code>	RAM usage	<code>free -h</code>
<code>df -h</code>	Disk usage	<code>df -h</code>

Example (try it in your terminal):

```
$ uname -a
Linux ubuntu 6.8.0-45-generic #45-Ubuntu SMP x86_64 GNU/Linux
$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04 LTS"
$ free -h
              total    used    free
Mem:          7.7Gi  2.1Gi  4.2Gi
$ df -h /
Filesystem Size  Used Avail Use% Mounted on
/dev/sda1  98G   31G   62G   34% /
```

6 Process Commands

Command	Use	Example
<code>ps aux</code>	List running processes	<code>ps aux</code>
<code>top</code>	Monitor processes live (press q to quit)	<code>top</code>
<code>pgrep</code>	Find process ID	<code>pgrep firefox</code>
<code>kill</code>	Stop a process	<code>kill 1234</code>

Example (try it in your terminal):

```
$ pgrep firefox
2841
$ ps aux | grep firefox
student 2841  4.2  6.1  ... /usr/lib/firefox/firefox
$ kill 2841
```

Security note: Unknown or suspicious processes in `ps aux` (odd names, very high CPU, running from `/tmp`) are a common sign that something is wrong.

7 Networking Commands

Command	Use	Example
<code>ip addr</code>	Show IP addresses	<code>ip addr</code>
<code>ip route</code>	Show routing table	<code>ip route</code>
<code>ping</code>	Test connectivity	<code>ping 8.8.8.8</code>
<code>ss -tuln</code>	Show listening ports	<code>ss -tuln</code>
<code>hostname</code>	Show system hostname	<code>hostname</code>

Example (try it in your terminal):

```
$ ip addr
2: eth0: <BROADCAST,MULTICAST,UP> ...
    inet 192.168.1.25/24 brd 192.168.1.255 scope global eth0
$ ping -c 3 8.8.8.8
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=14.2 ms
$ ss -tuln
Netid State  Local Address:Port
tcp    LISTEN  0.0.0.0:22
```

Security note: `ss -tuln` shows which ports your machine is listening on. Every open port is a possible entry point, so know what each one is for.

8 Search Commands

Command	Use	Example
<code>find</code>	Find files / directories	<code>find /home -name "*.txt"</code>
<code>grep</code>	Search text	<code>grep "password" file.txt</code>
<code>locate</code>	Search filenames	<code>locate file.txt</code>

Example (try it in your terminal):

```
$ find /home -name "*.txt"
/home/student/cybersecurity/notes.txt
$ grep "password" file.txt
password=admin123
$ grep -i "failed" /var/log/auth.log
Sep 30 09:14:02 sshd[911]: Failed password for root from 10.0.0.8
```

Security note: `grep` on log files is one of the most useful skills for spotting failed logins and suspicious activity.

9 Logs

Important cybersecurity location: `/var/log/`

Example (try it in your terminal):

```
$ ls /var/log
auth.log  syslog  kern.log  dpkg.log
$ cat /var/log/syslog
```

On some distributions, authentication logs are here:

```
$ cat /var/log/auth.log      (Debian / Ubuntu)
$ cat /var/log/secure       (RHEL / Fedora)
$ tail -f /var/log/auth.log (watch new entries live)
```

10 Package Management

Ubuntu / Debian:

```
$ sudo apt update
$ sudo apt install nmap
$ sudo apt remove nmap
```

Fedora / RHEL:

```
$ sudo dnf install nmap
$ sudo dnf remove nmap
```

Security note: Only install software from trusted repositories, and keep the system updated (apt update / apt upgrade) to get security fixes.

11 Text Processing

Command	Use	Example
head	First 10 lines	head file.txt
tail	Last 10 lines	tail file.txt
wc	Count lines / words / bytes	wc file.txt
sort	Sort lines	sort file.txt
uniq	Remove repeated lines (use after sort)	uniq file.txt

Example (try it in your terminal):

```
$ cat users.txt
ravi
anu
ravi
$ cat users.txt | sort | uniq
anu
ravi
$ wc -l users.txt
3 users.txt
```

12 Pipes and Redirection

Command	Use	Example
>	Write / overwrite	echo "Hello" > file.txt
>>	Append	echo "Linux" >> file.txt
 	Pipe output to another command	ls -l grep ".txt"

Example (try it in your terminal):

```
$ echo "Hello" > file.txt
$ echo "Linux" >> file.txt
$ cat file.txt
Hello
Linux
$ ls -l | grep ".txt"
-rw-r--r-- 1 student student 12 Sep 30 09:30 file.txt
```

Security note: A single > overwrites the file without warning. Use >> when you want to add to it.

13 Help Commands

Command	Use	Example
<code>man ls</code>	Manual page for a command	<code>man chmod</code>
<code>ls --help</code>	Quick help / options	<code>ls --help</code>
<code>history</code>	Commands you ran earlier	<code>history</code>
<code>clear</code>	Clear the screen	<code>clear</code>

Example (try it in your terminal):

```
$ man chmod          (shows the manual for chmod - press q to quit)
$ history | tail -3
101 cd cybersecurity
102 ls -l
103 chmod 600 secret.txt
```

14 Linux Directory Structure (Cybersecurity View)

Everything in Linux lives under / (root). Knowing what each directory holds tells you where to look for configuration, users, logs and running processes.

Directory	What it holds
/etc	Configuration
/home	Users
/root	Root user
/var/log	Logs
/tmp	Temporary
/proc	Processes
/sys	System / hardware
/dev	Devices
/boot	Boot
/usr	Software
/opt	Optional software
/run	Runtime

Tip: the last page contains a photo of this table from class.

14.1 /var - Variable Data

/var contains data that changes frequently while the Linux system is running. Important directories include:

```
/var/log    -> System and application logs
/var/cache  -> Cached data
/var/tmp    -> Temporary files that may persist longer
/var/spool  -> Queued data, such as mail or print jobs
```

To view it:

```
$ ls /var
backups  cache  lib  local  lock  log  mail  opt  run  spool  tmp
```

Security note: /var/log is especially important because security and system events are commonly recorded there.

In simple words: /var = changing (variable) system data, especially logs and caches.

14.2 /tmp - Temporary Files

/tmp stores temporary files and data created by applications and users while the system is running. Examples: temporary files created by applications, temporary installation files, short-lived working files.

View it with:

```
$ ls /tmp
ssh-XXXXXXabc  systemd-private-1a2b3c  test.txt
```

In simple words: /tmp = temporary storage for files that are not meant to be permanent.

14.3 /proc - Process & Kernel Information

/proc is a virtual filesystem that provides information about running processes and the Linux kernel.

```
/proc/cpuinfo    -> CPU information
/proc/meminfo    -> Memory information
/proc/version    -> Kernel version
/proc/<PID>/      -> Information about a specific process
```

Useful commands:

```
$ cat /proc/cpuinfo
model name : Intel(R) Core(TM) i5 ...
$ cat /proc/meminfo
MemTotal:      8054000 kB
$ cat /proc/version
Linux version 6.8.0-45-generic ...
```

Security note: /proc can help you inspect running processes, system resources and kernel information.

In simple words: /proc = process and kernel information.

15 Commands to Practice First for Cybersecurity

pwd	ls	cd	mkdir	touch	cat
cp	mv	rm	whoami	id	who
chmod	chown	uname	ps	top	ip
ss	ping	find	grep	history	sudo

Don't just memorize them - open a Linux terminal and type each command yourself.

Class Reference Photo - Linux Directory Slide

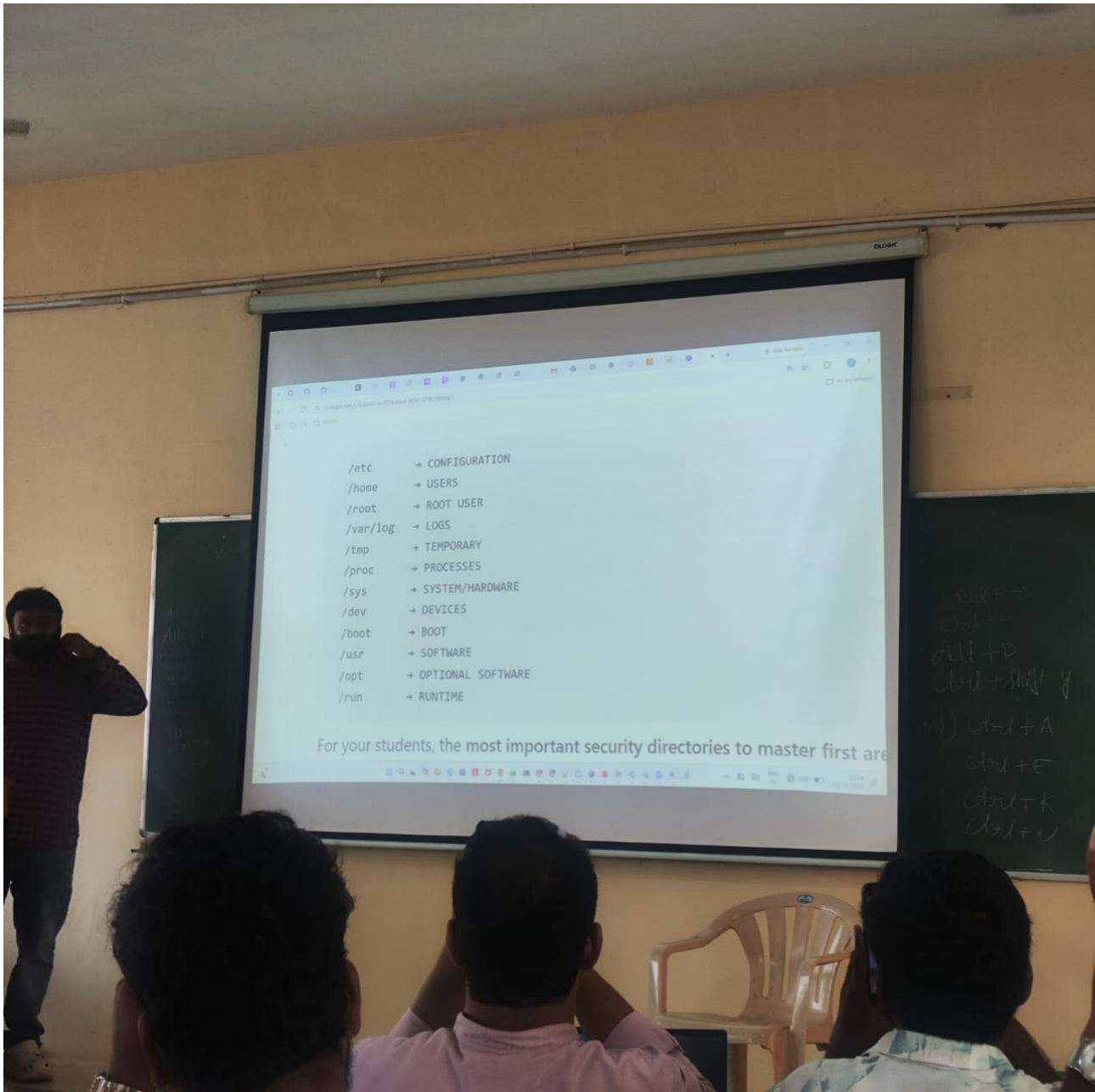


Photo of the slide shown in class, listing key Linux directories: /etc, /home, /root, /var/log, /tmp, /proc, /sys, /dev, /boot, /usr, /opt and /run.